Functional Connectivity Reconfiguration in N-back Working-Memory

By: Arefeh Lali Dehaghi, Goutham Arcod, Jaime Pineda, Kerem Akyurt, Valeria Moraga
Ifrit Ras el Hanout - The Gammas

neuromatch academy

Meet our team!

Arefeh Lali Dehaghi
Actor

Valeria Moraga
Doctor

Jaime Pineda
Chef

Kerem Akyurt
Accountant

Goutham Arcod
Journalist

Does load-related brain connectivity predict working-memory performance?

We predicted two things:

Pattern

A 78-feature FC fingerprint of the 2-back - 0-back change predicts performance.

Distributed pattern

Direction

Higher load shifts networks toward integration; larger shifts accompany better performance.

One-number direction

A distributed pattern and a one-number direction are different claims

We evaluated a 78-feature FC difference in held-out people

FC reconfiguration = how much each network pair changes its coupling when memory load goes from 0-back to 2-back.

1. Cohort

336 participants
HCP N-back dataset
2 WM runs per subject
360 Glasser ROIs
TR = 0.72 s

2. Condition frames

4 s HRF shift window
312 frames per load
No 0-back / 2-back temporal overlap

3. FC per condition

360 × 360 Pearson correlation matrix
Computed separately for 0-back and 2-back

4. Network fingerprint

78 total values over Cole-Anticevic networks:
- 12 within-network
- 66 between-network

5. Reconfiguration

fingerprint(2-back) - fingerprint(0-back)
Resulting feature matrix:
336 participants × 78 features

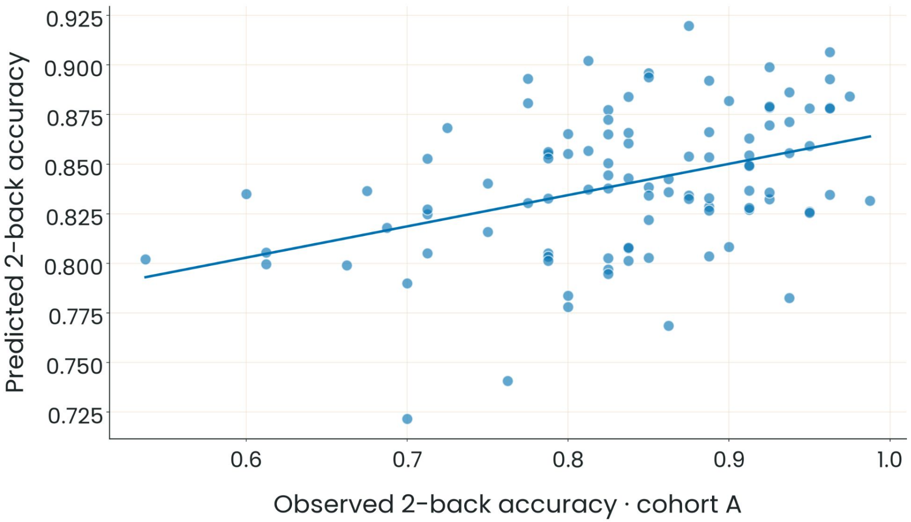
6. Model pipeline

StandardScaler + RidgeCV
Cross-validated regression model.
Fitted strictly inside each training fold.

7. Held-out evaluation

Repeated 5-fold CV • Fixed holdout partition • 1000-permutation null distribution • **B → A transfer** (301 → 100, identity-disjoint)
All splits hold out participants; “±” represents split SD, not CI.

The FC pattern predicted performance and transferred to a separate cohort



B → A Transfer Result

r = 0.398

Bootstrap 95% CI [0.25, 0.53]

301 B-only → 100 A (35 shared identities removed)

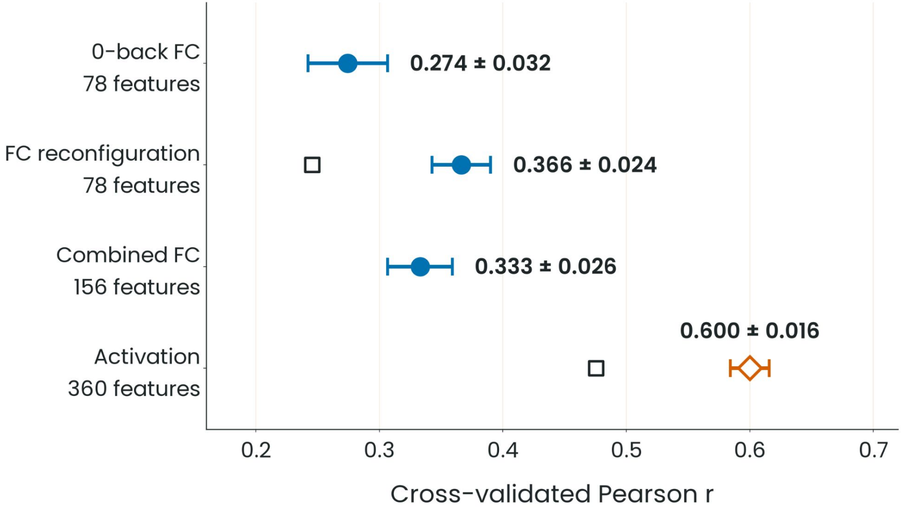
Primary Repeated CV

r = 0.366 ± 0.024

336 participants · 78 FC features

- *Identity-disjoint same-HCP transfer—not independent-site validation.*
- *SD summarizes sensitivity across 20 overlapping partitions, not population uncertainty.*

The prediction held—but a simpler regional signal predicted better



Repeated-CV Correlation (mean ± split SD)

0-back FC: **0.274 ± 0.032**

FC reconfiguration: **0.366 ± 0.024**

0-back + reconfiguration: **0.333 ± 0.026**

Activation contrast: **0.600 ± 0.016**

Direction, as predicted—but only at group level

Segregation fell under load (0.3271 → 0.3035; Δ = −0.0236; $p = 3.45 \times 10^{-5}$), yet larger shifts did not predict better performance ($r = -0.105$; $p = .054$).

Methodological Caveats & Guardrails

- **No biological winner claimed:** Unmatched features (360 activation vs 78 network FC). Activation does not erase FC findings but challenges specificity.
- **Not load-specific:** 0-back activation alone predicts as well (0.571), indicating a general benchmark.
- **Controls:** Activation is mean BOLD difference (not GLM beta). No cerebrovascular reactivity (CVR) control exists.
- **Open squares:** Held-out cross-run generalization.

Predictive signal survives; connectivity-specific mechanism remains unresolved

Survives

The pattern hypothesis

- **A 78-feature FC difference** predicts unseen 2-back accuracy.
- **The model transfers** across identity-disjoint same-HCP cohorts.

Refined

The directional hypothesis

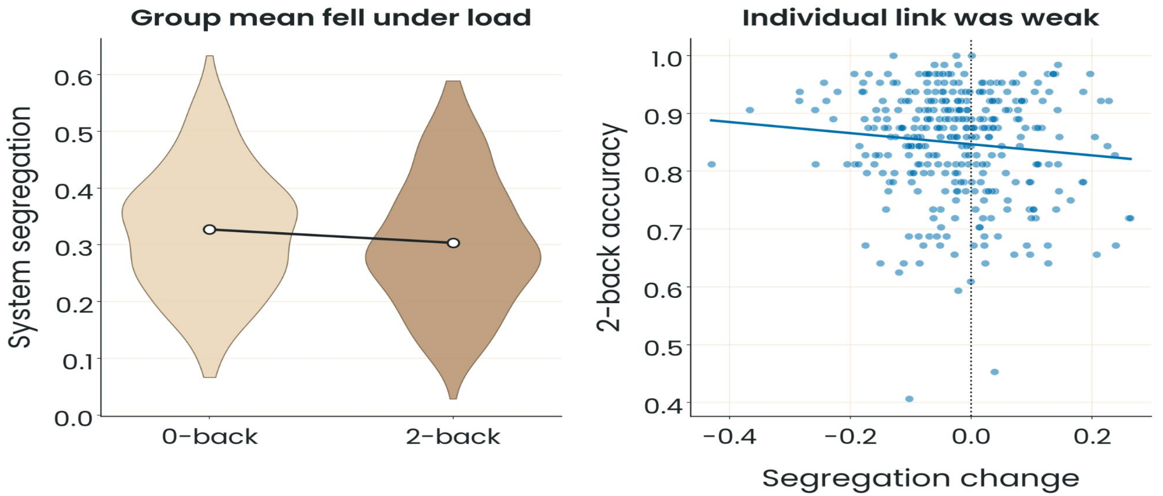
- **Segregation fell** under load, but larger shifts did not predict better performance.
- **Reconfiguration** showed no clear gain beyond 0-back FC.
- **FC added no clear gain** over activation under the current unmatched comparison.

Unresolved

Mechanistic open questions

- **Is the predictive info** connectivity-specific, or shared with task activation?
- **Vascular reactivity (CVR)** is not controlled in the activation benchmark.

Group direction was real; the individual link was weak



Statistical Summary

Group mean: 0-back 0.3271 → 2-back 0.3035

Paired change: Δ = -0.0236 ($p = 3.45 \times 10^{-5}$)

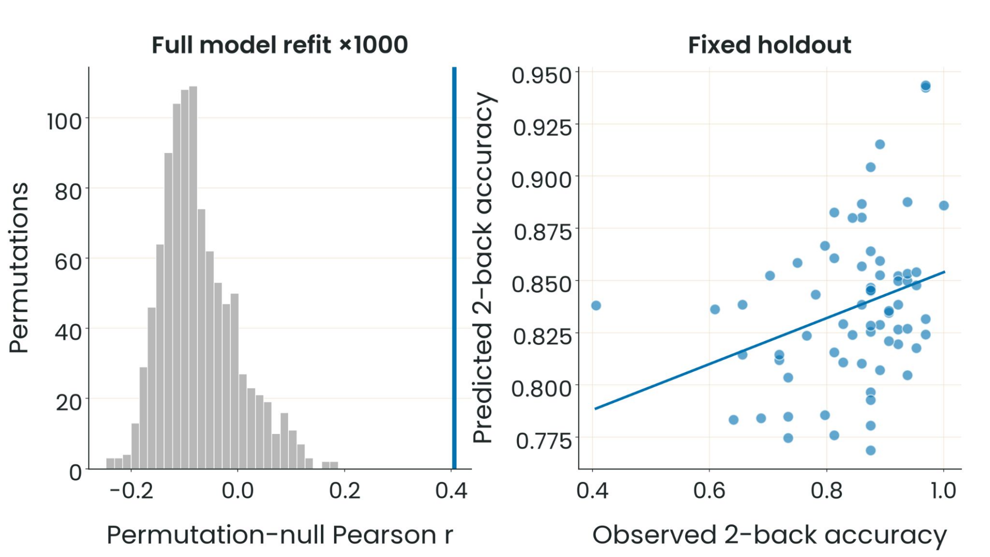
Across participants: r = -0.105 (p = .054)

Key Takeaways & Caveats

A reliable group-level mean shift does not establish individual predictive relevance.

Methodological limit: System segregation is a Chan-style scalar summary, not a replacement for the 78-feature multivariate pattern.

The checks behind the primary result



Validation Details

Fixed holdout: r = 0.312 in 67 unseen participants

Full-refit null (seed 42): r = 0.405; p = 1/1001 ≈ .001

B→A A-label permutation: p = 1/1001 ≈ .001

Reconfiguration over 0-back FC: $\Delta R^2 = +0.0344 \pm 0.0225$ → no clear gain

FC over activation: $\Delta R^2 = -0.0030 \pm 0.0065$ → no clear gain

Methodological Notes & Caveats:

The permutation p-value belongs only to seed-42 r = 0.405; the holdout is a separate split. The 2-SD decision rule is a split-sensitivity heuristic, not a formal superiority or equivalence test.

Four tests could resolve the remaining question

01 Match activation and FC dimensionality in nested participant-level CV.

02 Separate task coactivation from coupling with a prespecified estimator.

03 Test independent-site, repeat-session, family-aware generalization.

04 Run the activation model on resting-state amplitude: if rest predicts 2-back accuracy as well as task frames, the signal is task-independent.

Decision Criterion

FC must add reliable held-out value beyond activation and single-condition FC.

Note: These represent proposed validation experiments rather than completed research pipeline analyses.

References and statistical guardrails

Key Literature & References

Avery et al. (2020)

Distributed patterns of FC predict WM · J. Cognitive Neuroscience · PMC8004893

Chan et al. (2014)

Decreased segregation of brain systems · PNAS

Murphy et al. (2020)

Multimodal network dynamics underpinning WM · Nature Communications · 10.1038/s41467-020-15541-0

Masharipov et al. (2024)

Task-modulated FC methods · Communications Biology · 10.1038/s42003-024-07088-3

Hedge et al. (2018)

The reliability paradox · Behavior Research Methods · 10.3758/s13428-017-0935-1

Logothetis (2008)

What we can do and what we cannot do with fMRI · Nature

Full annotated bibliography: manuscript/references.md

Statistical Guardrails & Definitions

$r = 0.366 \pm 0.024$

Mean ± split SD across 20 repeated-CV partitions.

seed-42 $r = 0.405$; $p \approx .001$ (1/1001)

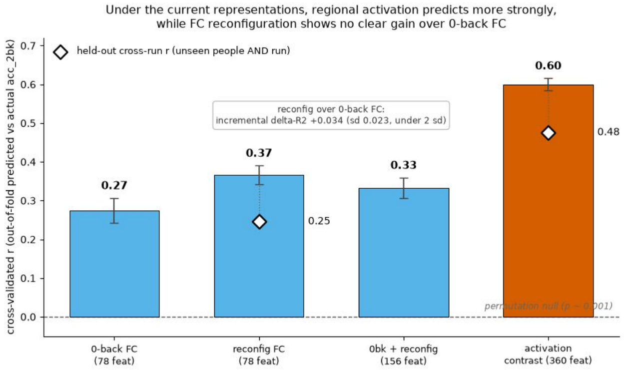
Full-refit permutation baseline.

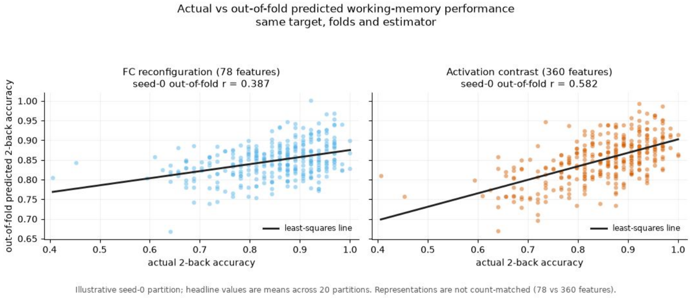
B→A $r = 0.398$; 95% CI [0.25, 0.53]

Identity-disjoint same-HCP transfer metric.

Scientific Caveat: *References frame and contextualize limits; they do not establish exact replications or causal validation. Logothetis (2008) frames the CVR vascular reactivity gap as an external cited limitation rather than our own direct finding.*

FC Reconfiguration





Explanation¿¿¿

Point 1

Our study aims to...

Point 2

.....

So... What does this mean?

- Conclusion 1
- Conclusion 2
- etc

Limitations:

- L1
- L2

IMPORTANT CONCEPTS

FC reconfiguration still predicts performance and the result generalizes to a held-out cohort. However, it substantially changes our conclusion because reconfiguration does not clearly add unique information beyond single-condition FC, and task activation predicts performance even better

References

1. Shine JM, Bissett PG, Bell PT, Koyejo O, Balsters JH, Gorgolewski KJ, Moodie CA, Poldrack RA. The Dynamics of Functional Brain Networks: Integrated Network States during Cognitive Task Performance. Neuron. 2016 Oct 19;92(2):544-554. doi: 10.1016/j.neuron.2016.09.018. Epub 2016 Sep 29. PMID: 27693256; PMCID: PMC5073034.

2. Finc K, Bonna K, He X, Lydon-Staley DM, Kühn S, Duch W, Bassett DS. Dynamic reconfiguration of functional brain networks during working memory training. Nat Commun. 2020 May 15;11(1):2435. doi: 10.1038/s41467-020-15631-z. Erratum in: Nat Commun. 2020 Jul 30;11(1):3891. doi: 10.1038/s41467-020-17747-8. PMID: 32415206; PMCID: PMC7229188.

3. Zhang H, Meng C, Di X, Wu X, Biswal B. Static and dynamic functional connectome reveals reconfiguration profiles of whole-brain network across cognitive states. Netw Neurosci. 2023 Oct 1;7(3):1034-1050. doi: 10.1162/netn_a_00314. PMID: 37781145; PMCID: PMC10473282.

4. Shen X, Finn ES, Scheinost D, Rosenberg MD, Chun MM, Papademetris X, Constable RT. Using connectome-based predictive modeling to predict individual behavior from brain connectivity. Nat Protoc. 2017 Mar;12(3):506-518. doi: 10.1038/nprot.2016.178. Epub 2017 Feb 9. PMID: 28182017; PMCID: PMC5526681.

5. Ray KL, Ragland JD, MacDonald AW, Gold JM, Silverstein SM, Barch DM, Carter CS. Dynamic reorganization of the frontal parietal network during cognitive control and episodic memory. Cogn Affect Behav Neurosci. 2020 Feb;20(1):76-90. doi: 10.3758/s13415-019-00753-9. PMID: 31811557; PMCID: PMC7018593.

Complementary Slides

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Complementary Slides

INTERNAL SLIDE: Compression map

15-Slide Internal-Review Version → Final Presentation Deck

Spoken Slides (Deck 1)

Slide 3: Absorbs Question and both hypotheses into one opening beat.

Slide 4: Cohorts, feature construction, and evaluation mapped to a pipeline schematic.

Slide 5: Focuses on Primary CV and transfer (sub-checks demoted to backup Slide 9).

Slide 6: Directional refinement & activation benchmark (segregation figure demoted to backup 8).

Slide 7: Conclusion, explicitly naming both opening hypotheses.

Backup & Omitted Assets

Backup Slide 8: Directional figure and full statistics.

Backup Slide 9: Null, fixed holdout, and incremental test.

Backup Slide 10: Future work, expanded to four tests.

Omitted Assets:

- Internal Slide 10 (anatomical context) stays out of both decks.
- Several assets remain unassigned in the chart library: condition-fc-contrast, anatomical-context, feature-construction, primary-repeated-cv, and incremental-fc-test.